Cohomological Stability of Rational Maps to a Moduli Space (Draft)

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Introduction

In this article, we investigate the cohomological stability of a 'nice component' of $\mathrm{Hom}_k(\mathbb{P}^1, M)$, the scheme of algebraic maps of degree k from $\mathbb{P}^1$ to M, where M is a moduli space. Our study focuses primarily on the case of odd degrees. We proceed as follows: First, we construct the moduli space via extensions of sheaves, following the work of Castravet [1]. Second, we analyze the unstable locus and identify it as an e-secant variety. Furthermore, we establish a stratification of singularities for this unstable locus. Subsequently, we analyze the geometry of the space of the rational lines intersecting the unstable locus and compute the dimension of this space. Finally, we apply established cohomological stability arguments, aiming to relate the cohomology of our component to known results through the Projective Bundle Formula and a sequence of birational maps. A strategy for analyzing these birational maps is outlined in the final part, providing a clear pathway toward a possible solution of the stability conjecture.

1 Problems

In this article we want to study the cohomological stability of a nice component of $\mathrm{Hom}_k(\mathbb{P}^1, M)$, the scheme of algebraic maps of degree k from $\mathbb{P}^1$ to M, where M is a moduli space.

Let C be a genus $g \geq 2$ smooth projective curve over $\mathbb{C}$. Let ξ be a degree 1 line bundle on C and let M_ξ be the moduli space of isomorphism classes of stable, rank 2 vector bundles $\mathcal{E}$ on C, with determinant $\det(\mathcal{E}) \cong \bigwedge^2(\mathcal{E}) \cong \xi$. Then M_ξ is a smooth, projective, irreducible variety of dimension $(3g-3)$. It is known that $\mathrm{Pic}(M_\xi) \cong \mathbb{Z}$ and let Θ be the ample generator. By a rational curve of degree k on M_ξ we will always mean a non-constant morphism $f : \mathbb{P}^1 \to M_\xi$, such that the line bundle $f^*\Theta$ on $\mathbb{P}^1$ has degree k. We first construct a 'nice component' of the Hilbert scheme $\mathrm{Hom}_k(\mathbb{P}^1, M_\xi)$ of morphisms $f : \mathbb{P}^1 \to M_\xi$ of degree k by extensions of sheaves.

We first consider the odd case. Let e be an integer and let $\mathcal{L}$ be a line bundle on C of degree $-e$. Consider extensions:

$$0 \to \mathcal{L} \to \mathcal{E} \to \mathcal{L}^{-1} \otimes \xi \to 0. \quad (*)$$

Then $\mathcal{E}$ is a rank 2 vector bundle and such extensions are classified by the vector space

$$V_{\mathcal{L}} = \mathrm{Ext}^1_C(\mathcal{L}^{-1} \otimes \xi, \mathcal{L}) \cong H^1(C, \mathcal{L}^2 \otimes \xi^{-1})$$

Using Riemann-Roch, the vector space $V_{\mathcal{L}}$ has dimension

$$\dim V_{\mathcal{L}} = -\deg(\mathcal{L}^2 \otimes \xi^{-1}) - 1 + g = 2e + g$$

And the isomorphism classes of non-trivial extensions as above are parametrized by the projective space $\mathbb{P}(V_{\mathcal{L}}) = \mathrm{Proj}(\mathrm{Sym}V_{\mathcal{L}}^*)$.

We now let $\mathcal{L}$ vary in the Picard variety $\mathrm{Pic}^{-e}(C)$ of line bundles on C of degree $-e$ and define a global parameter space for extensions of type (*). Let $\mathcal{A}$ be a Poincaré bundle on $\mathrm{Pic}^{-e}(C) \times C$ and let π_1, π_2 be the two projections from $\mathrm{Pic}^{-e}(C) \times C$. Define on $\mathrm{Pic}^{-e}(C)$ the relative extension sheaf

$$\mathcal{S} = \mathcal{E}xt^1_{\mathrm{Pic}^{-e}(C)\times C|\mathrm{Pic}^{-e}(C)}(\mathcal{A}^{-1} \otimes \pi_2^*\xi, \mathcal{A})$$

Note that $\mathcal{S}$ is a locally free sheaf. Let X be the projective bundle $\mathbb{P}(\mathcal{S})$ and let

$$p : X \to \mathrm{Pic}^{-e}(C)$$

$$p_C : X \times C \to \mathrm{Pic}^{-e}(C) \times C$$

be the corresponding maps. Let ν_1, ν_2 be the two projections from $X \times C$. There is a universal extension on $X \times C$:

$$0 \to \nu_1^*\mathcal{O}_X(1) \otimes p_C^*\mathcal{A} \to \mathcal{G} \to p_C^*(\mathcal{A}^{-1}) \otimes \nu_2^*\xi \to 0. \quad (1.1)$$

It is universal in the sense that, if $x \in X$ and we let $\mathcal{L} = p(x) \in \mathrm{Pic}^{-e}(C)$, when we restrict (1.1) to $\{x\} \times C$, we get an exact sequence:

$$0 \to \mathcal{L} \to \mathcal{G}_x \to \mathcal{L}^{-1} \otimes \xi \to 0 \quad (1.2)$$

whose class in $\mathbb{P}(V_{\mathcal{L}}) \cong p^{-1}(\{\mathcal{L}\})$ is x.

We define $X_{\mathcal{L}} = \mathbb{P}(V_{\mathcal{L}})$ and $Z_{\mathcal{L}} \subset X_{\mathcal{L}}$ is the unstable locus. Then X is the union of all $X_{\mathcal{L}}$ as $\mathcal{L}$ varies in $\mathrm{Pic}^{-e}(C)$, and we define Z as the union of all $Z_{\mathcal{L}}$. Let $M = \mathrm{Hom}_1(\mathbb{P}^1, X \setminus Z)$, then it is known that an element in M is a rational curve on M of degree $2e+1$, which can be regarded as an element in $\mathrm{Hom}_{2e+1}(\mathbb{P}^1, M)$. Furthermore, the closure of the image of M in $\mathrm{Hom}_{2e+1}(\mathbb{P}^1, M_\xi)$ is an irreducible component with expected dimension $2(2e+1) + 3g - 3$. That explains why it is called 'nice component'. We can also construct 'nice component' for the even case, but instead we need to use extension

$$0 \to \mathcal{E} \to \mathcal{E}' \to \mathcal{O}_D \to 0$$

where $D \in \mathrm{Sym}^e(C)$ and let $\mathcal{E}$ be a rank 2, semistable vector bundle on C, with $\det(\mathcal{E}) \cong \xi(-D)$. But in this draft, we mainly consider the odd cases.

Now we want to study the cohomology of M as the degree grows. We have a conjecture:

Conjecture 1.1. *The cohomology of M stabilizes as the degree goes to infinity.*

One explanation of this conjecture is that the spaces $\mathrm{Hom}_k(\mathbb{P}^1, M)$ are usually ideal candidates for cohomological stability, where M is a projective manifold. To be more specific, let $\mathcal{M}_k = C_k^\infty(\mathbb{P}^1, M)$, the smooth maps of topological degree k. And we define an energy functional on this space:

$$E(f) = \frac{1}{2}\int_{\mathbb{P}^1} |df|^2 \, d\,vol$$

Let $\mathcal{M}_k^{hol}$ be the space of all holomorphic maps from $\mathbb{P}^1$ to M, then M_k^{hol} is just the set of all global minima in $\mathcal{M}$ with respect to the energy functional E. According to Morse theory, the difference in topology between the space of holomorphic maps $\mathcal{M}_k^{hol}$ and the full space of smooth maps $\mathcal{M}_k$ is determined by the critical points of the energy functional with higher index (saddle points). It is known (or expected) that the Morse indices of these non-holomorphic critical points grow with the degree k. Consequently, the inclusion $\mathcal{M}_k^{hol} \hookrightarrow \mathcal{M}_k$ induces an isomorphism on cohomology in a range of degrees that increases with k:

$$\mathrm{H}^i(\mathcal{M}_k^{hol}) \simeq \mathrm{H}^i(\mathcal{M}_k), \text{for } i \leq C(k)$$

where $\lim_{k\to\infty} C(k) = \infty$. Hence, that explains why the cohomological stability may hold.

And our goal is to study whether cohomological stability also holds as M is a moduli space.

2 Analysis

We denote $N = \mathrm{Hom}_1(\mathbb{P}^1, X)$. And $Y = \{l \in N, |\ l \text{ meets } Z\}$, denotes the 'bad maps' inside N. And $M = N \setminus Y$, which is our desired 'nice component'. We can also define $N_{\mathcal{L}} = \mathrm{Hom}_1(\mathbb{P}^1, X_{\mathcal{L}})$, $Y_{\mathcal{L}} = \{l \in N_{\mathcal{L}}, |\ l \text{ meets } Z\}$ and $M_{\mathcal{L}} = N_{\mathcal{L}} \setminus Y_{\mathcal{L}}$

To understand the cohomology of M, we need to understand:

1. $\mathrm{H}^k(M_{\mathcal{L}})$, the cohomology of $M_{\mathcal{L}}$;
2. how the cohomology $\mathrm{H}^k(M_{\mathcal{L}})$ varies as $\mathcal{L}$ varies in $\mathrm{Pic}^e(C)$.

So we will first study the structure of $Y_{\mathcal{L}}$.

2.1 Structure of $Z_{\mathcal{L}} \subset X_{\mathcal{L}}$

We first analyze the unstable locus $Z_{\mathcal{L}}$. The bundle $\mathcal{E}$ in $(*)$ is not stable if and only if there exists a line bundle $\mathcal{L}'$ on C of degree 1 and a non-zero morphism $\mathcal{L}' \to \mathcal{E}$. Then the morphism $\mathcal{L}' \to \mathcal{L}^{-1} \otimes \xi$ is non-zero as well. This is because there is no non-zero morphism $\mathcal{L}' \to \mathcal{L}$ as

$$\deg(\mathcal{L}') = 1 > -e = \deg(\mathcal{L})$$

Therefore, it follows that there is some effective divisor D on C of degree e such that $\mathcal{L}' \cong \mathcal{L}^{-1} \otimes \xi(-D)$. Let $\mathcal{E}'$ be the kernel of the composition morphism $\mathcal{E} \to \mathcal{L}^{-1} \otimes \xi \to \mathcal{L}^{-1} \otimes \xi|_D$. There is a commutative diagram with the two horizontal sequences exact:

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{E}' \arrow[r] \arrow[d] & \mathcal{E} \arrow[r] \arrow[d] & \mathcal{L}^{-1} \otimes \xi|_D \arrow[r] \arrow[d, "\cong"] & 0 \\
0 \arrow[r] & \mathcal{L}^{-1} \otimes \xi(-D) \arrow[r] & \mathcal{L}^{-1} \otimes \xi \arrow[r] & \mathcal{L}^{-1} \otimes \xi|_D \arrow[r] & 0
\end{tikzcd}$$

Using the snake lemma, we get that there is an exact sequence

$$0 \to \mathcal{L} \to \mathcal{E}' \to \mathcal{L}^{-1} \otimes \xi(-D) \to 0 \tag{2.1}$$

Moreover, the following composition of morphisms is zero:

$$\mathcal{L}^{-1} \otimes \xi(-D) \cong \mathcal{L}' \to \mathcal{E} \to \mathcal{L}^{-1} \otimes \xi|_D$$

Then $\mathcal{L}^{-1} \otimes \xi(-D)$ must map to the subbundle $\mathcal{E}'$ and the exact sequence (2.1) is split.

We conclude that the vector $v \in V_{\mathcal{L}} = \mathrm{Ext}^1_C(\mathcal{L}^{-1} \otimes \xi, \mathcal{L})$, corresponding to an unstable extension, is in the kernel $\mathrm{Ext}^1_C(\mathcal{L}^{-1} \otimes \xi \otimes \mathcal{O}_D, \mathcal{L})$ of the surjective map,

$$\mathrm{Ext}^1_C(\mathcal{L}^{-1} \otimes \xi, \mathcal{L}) \to \mathrm{Ext}^1_C(\mathcal{L}^{-1}(-D) \otimes \xi, \mathcal{L})$$

Hence, the unstable locus is image of all $\mathrm{Ext}^1_C(\mathcal{L}^{-1} \otimes \xi \otimes \mathcal{O}_D, \mathcal{L}) \subset \mathrm{Ext}^1_C(\mathcal{L}^{-1} \otimes \xi, \mathcal{L})$ for all D, where D is an effective divisor of degree d, we can assume $D \in \mathrm{Sym}^e(C)$. To make the inclusion more clear, we take duality and use Serre duality,

$$\mathrm{Ext}^1_C(\mathcal{L}^{-1}\otimes\xi, \mathcal{L})^* \simeq H^0(C, \mathcal{L}^{-2}\otimes\xi\otimes\omega_C) \to H^0(C, \mathcal{L}^{-2}\otimes\xi\otimes\omega_C\otimes\mathcal{O}_D) \simeq \mathrm{Ext}^1_C(\mathcal{L}^{-1}\otimes\xi\otimes\mathcal{O}_D, \mathcal{L})^*$$

Let $\mathcal{K} = \mathcal{L}^{-2} \otimes \xi \otimes \omega_C$. This map means restricting a section of $\mathcal{K}$ onto linear combination of points of D. Hence, this map can be interpreted like this: by the sheaf $\mathcal{K}$ over C, we get an embedding $C \to \mathbb{P}(\mathrm{Ext}^1_C(\mathcal{L}^{-1} \otimes \xi, \mathcal{L})^*)$. (Here $\deg \mathcal{K} = 2e + 1 + 2g - 2 = 2e - 1 + 2g \geq 2g + 1$. Then $\mathcal{K}$ is very ample, hence it defines an embedding.) And the image of $\mathbb{P}(\mathrm{Ext}^1_C(\mathcal{L}^{-1} \otimes \xi \otimes \mathcal{O}_D, \mathcal{L}))$, is just the projective subspace generated by e points $P_1, P_2, \cdots, P_e$, where $D = P_1 + P_2 + \cdots + P_e$. Therefore, we understand the structure of $Z_{\mathcal{L}}$: $Z_{\mathcal{L}}$ **is the e-secant variety of curve** C**,** where C is embedded into $\mathbb{P}(V_{\mathcal{L}})$ by very ample line bundle $\mathcal{K}$. In general case, the dimension of $Z_{\mathcal{L}}$ is $2e - 1$.

We now study the singularity of $Z_{\mathcal{L}}$. We will prove the following proposition:

Proposition 2.1 (Singular Locus)**.** *For the curve $C \subset \mathbb{P}^N$ defined above, the singular locus of $Z_{\mathcal{L}}$ is exactly the $(e-1)$-secant variety:*

$$\mathrm{Sing}(Z_{\mathcal{L}}) = Sec_{e-1}(C).$$

Specifically, the singularities arise solely from the degeneration of the configuration of e points (i.e., when points coincide), transforming secant planes into tangent planes.

Proposition 2.1 can be easily deduced from the following proposition:

Proposition 2.2 (Linear Independence of $2e$ Points)**.** *Let $C \subset \mathbb{P}^N$ be the curve embedded by the line bundle $\mathcal{K}$ described above. For any collection of $k = 2e$ distinct points $D = x_1 + \cdots + x_{2e}$ on C, these points are linearly independent in $\mathbb{P}^N$.*

Proof. The points in the support of the divisor D are linearly independent if and only if they impose independent conditions on the linear system $|\mathcal{K}|$. Equivalently, we must satisfy the condition:

$$h^0(C, \mathcal{K}) - h^0(C, \mathcal{K}(-D)) = \deg(D) = 2e.$$

By

$$0 \longrightarrow \mathcal{K}(-D) \longrightarrow \mathcal{K} \longrightarrow \mathcal{K}|_D \longrightarrow 0$$

We get the long exact sequence of cohomology:

$$0 \to H^0(C, \mathcal{K}(-D)) \to H^0(C, \mathcal{K}) \to H^0(C, \mathcal{K}|_D) \to H^1(C, \mathcal{K}(-D)) \to H^1(C, \mathcal{K}) \to 0$$

Using the exact sequence of cohomology, this we only need to prove:

$$h^1(C, \mathcal{K}(-D)) = 0.$$

We compute this dimension using Serre Duality. By duality, we have:

$$h^1(C, \mathcal{K}(-D)) = h^0(C, \omega_C \otimes (\mathcal{K}(-D))^\vee) = h^0(C, \omega_C \otimes \mathcal{K}^{-1}(D)).$$

Let us compute the degree of the line bundle $\mathcal{M} = \omega_C \otimes \mathcal{K}^{-1}(D)$:

$$\begin{aligned}\deg(\mathcal{M}) &= \deg(\omega_C) - \deg(\mathcal{K}) + \deg(D) \\ &= (2g-2) - (2e+2g-1) + 2e \\ &= 2g-2-2e-2g+1+2e \\ &= -1.\end{aligned}$$

Since $\deg(\mathcal{M}) = -1 < 0$, the line bundle $\mathcal{M}$ admits no non-zero global sections. Therefore:

$$h^0(C, \mathcal{M}) = 0 \implies h^1(C, \mathcal{K}(-D)) = 0.$$

This confirms that any $2e$ points on the curve impose independent linear conditions. $\square$

Therefore, the unstable locus $Z_{\mathcal{L}}$ possesses a canonical stratification of singularities:

$$Z_{\mathcal{L}} = Sec_e(C) \supset Sec_{e-1}(C) \supset \cdots \supset C.$$

2.2 Dimension Analysis of $Y_{\mathcal{L}}$

To analyze the structure of the bad maps $Y_{\mathcal{L}}$, we first understand its geometry via an incidence correspondence. Recall that $Z_{\mathcal{L}} \subset X_{\mathcal{L}} \cong \mathbb{P}^N$ is the unstable locus, identified as the e-secant variety of the curve C embedded by the line bundle $\mathcal{K}$. Let $N_{\mathcal{L}} = \mathbb{G}(1, \mathbb{P}^N)$ be the Grassmannian of line. Here we need to clarify: the $\mathbb{G}(1, n)$ is different from $\mathrm{Hom}_1(\mathbb{P}^1, \mathbb{P}^n)$, since they differ by $\mathrm{Aut}(\mathbb{P}^1) \simeq \mathrm{PGL}(2)$. But in order to make the discussion easier, we do not consider the action of $\mathrm{Aut}(\mathbb{P}^1)$ in section 2. We do not consider the $\mathrm{PGL}(2)$ action on $Y_{\mathcal{L}}$, $M_{\mathcal{L}}$ either.

We define the incidence variety $\mathcal{I}$ as:

$$\mathcal{I} = \{(\ell, p) \in N_{\mathcal{L}} \times Z_{\mathcal{L}} \mid p \in \ell\}$$

We have two natural projections: $q : \mathcal{I} \to Z_{\mathcal{L}}$ and $\pi : \mathcal{I} \to Y_{\mathcal{L}} \subset N_{\mathcal{L}}$. The map q makes $\mathcal{I}$ a $\mathbb{P}^{N-1}$-bundle over $Z_{\mathcal{L}}$, since the space of lines passing through a point $p \in \mathbb{P}^N$ is isomorphic to $\mathbb{P}^{N-1}$.

The dimension of the ambient projective space is $N = 2e + g - 1$. The dimension of the e-secant variety is generally $\dim Z_{\mathcal{L}} = 2e - 1$. Hence, the dimension of the incidence variety is $\dim \mathcal{I} = \dim Z_{\mathcal{L}} + (N-1) = (2e-1) + (2e+g-2) = 4e+g-3$. Since π is generically finite onto its image $Y_{\mathcal{L}}$, we have $\dim Y_{\mathcal{L}} = 4e+g-3$. Besides, the dimension of the total space $\dim N_{\mathcal{L}} = 2(N-1) = 4e+2g-4$, we get the codimension:

$$\mathrm{codim} Y_{\mathcal{L}} = (4e+2g-4) - (4e+g-3) = g-1.$$

For $g \geq 2$, $Y_{\mathcal{L}}$ is a proper subvariety of high codimension, which is consistent with the expectation that $M_{\mathcal{L}}$ is the "nice" generic component.

2.3 Some Cohomological Stability Arguments

We now focus on Conjecture 1.1. The strategy relies on relating the cohomology of the open set $M_{\mathcal{L}}$ to the homology of the closed set $Y_{\mathcal{L}}$ using the long exact sequence of the pair $(N_{\mathcal{L}}, M_{\mathcal{L}})$:

$$\cdots \to H^k(N_{\mathcal{L}}) \to H^k(M_{\mathcal{L}}) \to H^{k+1}(N_{\mathcal{L}}, M_{\mathcal{L}}) \to \cdots$$

Since $N_{\mathcal{L}}$ is a smooth complex manifold of dimension $D = 4e+2g-4$, we apply Alexander duality:

$$H^{k+1}(N_{\mathcal{L}}, M_{\mathcal{L}}; \mathbb{Q}) \cong H_{2D-k-1}(Y_{\mathcal{L}}; \mathbb{Q}).$$

Since $H^k(N_{\mathcal{L}})$ is stable (as the cohomology of a Grassmannian), the stability of $H^k(M_{\mathcal{L}})$ depends entirely on the stability of the $H_{2D-k-1}(Y_{\mathcal{L}})$.

Since the unstable locus $Z_{\mathcal{L}}$ is singular, we analyze its cohomology by passing to a smooth resolution. We consider the abstract e-secant variety $\tilde{Z}_{\mathcal{L}}$. The projection $\pi : \tilde{Z}_{\mathcal{L}} \to Sym^e(C)$ gives $\tilde{Z}_{\mathcal{L}}$ the structure of a projective bundle of rank $e-1$. Let $\rho : \tilde{Z}_{\mathcal{L}} \to Z_{\mathcal{L}} \subset \mathbb{P}(V_{\mathcal{L}})$ be the natural proper surjective morphism mapping the abstract secant variety to the embedded secant variety.

To understand their cohomology, we need the following three key theorems.

Theorem 2.3 (Projective Bundle Formula / Leray-Hirsch)**.** *Let $E \xrightarrow{\pi} B$ be a projective bundle of rank r. The Poincaré polynomial of the total space factorizes as:*

$$P_E(t) = P_B(t) \cdot P_{\mathbb{P}^r}(t) = P_B(t) \cdot (1 + t^2 + \cdots + t^{2r}).$$

Theorem 2.4 (Macdonald, 1962 [2])**.** *The cohomology of the symmetric product $Sym^e(C)$ stabilizes. Specifically, the Betti numbers $b_q(Sym^e(C))$ are independent of e for $0 \leq q < e$. Moreover, the cohomology ring is generated by the cohomology of the Jacobian $J(C)$.*

We now apply these theorems to understand the homology of $Y_{\mathcal{L}}$. Recall from the Alexander duality argument that the term of interest is $H_{2D-k-1}(Y_{\mathcal{L}})$, where D is the dimension of the ambient Grassmannian. As $e \to \infty$, the dimension D grows, so we need to consider the homology of $Y_{\mathcal{L}}$ in degrees close to the top dimension (co-degree $k+1$).

Since $Sym^e(C)$ is a smooth variety of complex dimension e, Poincaré Duality implies $b_{2e-q}(Sym^e(C)) = b_q(Sym^e(C))$. Macdonald's result ensures b_q is stable for fixed q. Consequently, the Betti numbers of $Sym^e(C)$ are stable in fixed co-degrees (degrees relative to the top dimension).

Applying the Projective Bundle Formula to $\tilde{Z}_{\mathcal{L}} \to Sym^e(C)$, the cohomology of $\tilde{Z}_{\mathcal{L}}$ is a sum of shifted copies of $H^*(Sym^e(C))$. Since the base exhibits cohomological stability, the total space $\tilde{Z}_{\mathcal{L}}$ also exhibits stability. And we have proper, surjective map $p : \tilde{Z}_{\mathcal{L}} \to Z_{\mathcal{L}}$. Also, we know $\mathcal{I}$ is a projective bundle over $Z_{\mathcal{L}}$, so we can get the cohomology from the cohomology of $Z_{\mathcal{L}}$ by Projective Bundle Formula. Also, we have proper, surjective map $\pi : \mathcal{I} \to Y_{\mathcal{L}}$. So now we need to make more specific calculations on π and p.

2.4 More Calculations

To determine the precise cohomology of $M_{\mathcal{L}}$, we must analyze the proper, surjective maps $p : \tilde{Z}_{\mathcal{L}} \to Z_{\mathcal{L}}$ and $\pi : \mathcal{I} \to Y_{\mathcal{L}}$. From the discussion in section 2.1, we first observe that p serves as a resolution of singularities and is an isomorphism over the dense open set $Z_{\mathcal{L}} \setminus \mathrm{Sec}_{e-1}(C)$. Similarly, since the codimension of $Z_{\mathcal{L}}$ is $g \geq 2$, a generic line in $\mathbb{P}^N$ intersects $Z_{\mathcal{L}}$ at a single point, implying that π is generically one-to-one; thus, both morphisms are birational. Since birational morphisms induce isomorphisms on cohomology modulo exceptional loci, our strategy is to explicitly compute the exceptional part for small values of e to understand their geometric structures. This part is the subject of ongoing work.

References

[1] A.-M. Castravet, *Rational families of vector bundles on curves, I*, arXiv preprint math/0302133 (2003).

[2] I. G. Macdonald, *Symmetric products of an algebraic curve*, *Topology I* (1962), 319–343.